

Operator-First Arithmetic (ii) as an Initiality Principle

Aim

Package **numerals-as-iterators** as an *initiality statement* (" $\langle s \rangle$ is the free monoid on one generator realized inside $\text{End}(N)$ "), and show how the usual **Natural Numbers Object (NNO)** story is recovered by adding a basepoint z and the recursion/universal property.

1. Pointed monoids and the free monoid on one generator

1.1 Pointed monoids

A **pointed monoid** is a pair (M, m) where M is a monoid and $m \in M$ is a distinguished element.

Let $\mathbf{Mon}_*$ be the category:

- **Objects:** pointed monoids (M, m) .
- **Morphisms:** monoid homomorphisms $h : M \rightarrow M'$ such that $h(m) = m'$.

1.2 Initiality (free on one generator)

The **free monoid on one generator** is the initial object in $\mathbf{Mon}_*$.

A standard choice is $(\mathbb{N}_0, +, 0; 1)$:

- underlying monoid $(\mathbb{N}_0, +, 0)$
- distinguished element 1

Universal property (initiality): For every pointed monoid (M, m) , there exists a unique pointed-monoid morphism

$$\phi_m : (\mathbb{N}_0, +, 0; 1) \rightarrow (M, \cdot, e; m)$$

with $\phi_m(1) = m$. Necessarily,

$$\phi_m(n) = m^n \quad \text{(the } n\text{-fold product of } m \text{ in } M\text{).}$$

Intuition: the “number” n is the operator “apply the generator n times.”

2.1 The endomorphism monoid

Fix an object N in a category $\mathcal{C}$. Its endomorphisms

$$\mathrm{End}(N) := \mathcal{C}(N, N)$$

form a monoid under composition ($\circ$) with unit id_N .

Pick a distinguished endomorphism

$$s: N \rightarrow N.$$

Then $(\mathrm{End}(N), s)$ is a pointed monoid.

2.2 The iterator homomorphism

By initiality of $(\mathbb{N}_0, +, 0; 1)$ in $\mathbf{Mon}_*$, there is a **unique** pointed-monoid morphism

$$\Phi_s: (\mathbb{N}_0, +, 0; 1) \rightarrow (\mathrm{End}(N), \circ, \mathrm{id}_N; s)$$

with $\Phi_s(1) = s$. Concretely:

$$\Phi_s(n) = s^n := \underbrace{s \circ s \circ \dots \circ s}_{n \text{ times}}, \quad \Phi_s(0) = \mathrm{id}_N.$$

Define the submonoid generated by s :

$$\langle s \rangle := \{s^n \mid n \in \mathbb{N}_0\} \subseteq \mathrm{End}(N).$$

Then:

- $\mathrm{im}(\Phi_s) = \langle s \rangle$
- $\langle s \rangle$ is a **realization of the free monoid on one generator** inside $\mathrm{End}(N)$ *exactly to the extent that* Φ_s is injective.

To ensure distinct numerals correspond to distinct iterators, impose:

$$s^m = s^n \implies m = n.$$

Equivalently, Φ_s is an embedding of monoids, so $\langle s \rangle \cong (\mathbb{N}_0, +, 0)$.

3. OFA(ii): numerals are iterators; values are induced

3.1 Definition (numerals as iterators)

In OFA(ii), a **numeral** is literally an endomorphism:

$$\widetilde{n} : N \rightarrow N$$

Arithmetic at the operator level is induced by the monoid laws:

- **Addition:** $\widetilde{m} + \widetilde{n} = \widetilde{m} \circ \widetilde{n}$.
- **Multiplication:** $\widetilde{m}\widetilde{n} = (\widetilde{m})^{\circ n} = s^{mn}$.

3.2 Induced “value at zero” map (recovering (i))

Add a basepoint (global element)

$$z : 1 \rightarrow N$$

Then each iterator yields a “value”:

$$\underline{n} := \widetilde{n} \circ z : 1 \rightarrow N$$

So (i) is an *evaluation* of (ii).

4.1 NNO universal property (recursion)

A **Natural Numbers Object** in $\mathcal{C}$ (with terminal object 1) is a triple (N, z, s) such that for every object A , every point $q : 1 \rightarrow A$, and every endomorphism $f : A \rightarrow A$, there exists a **unique** morphism $u : N \rightarrow A$ satisfying:

$$u \circ z = q, \quad u \circ s = f \circ u$$

This is the categorical form of primitive recursion/iteration.

When $\mathcal{C}$ has coproducts, the same structure can be packaged as an **initial algebra** for the endofunctor

$$F(X) = 1 + X$$

The structure map $\alpha : 1 + N \rightarrow N$ is determined by z and s , and initiality of (N, α) is equivalent to the recursion property above.

This is the cleanest “no extra normalization needed” route: uniqueness comes from initiality.

4.3 How recursion links back to iterators (OFA(ii) → semantics)

Given recursion data (A, q, f) , the unique $u : N \rightarrow A$ satisfies, for all n :

$$u \circ s^n = f^n \circ u$$

and in particular on the induced numerals:

$$u(\underline{n}) = u \circ s^n \circ z = f^n \circ u \circ z = f^n(q).$$

So **numerals-as-iterators** become the universal indexing of iterates of any process f starting from any seed q .

4.4 Recovering addition and multiplication (usual arithmetic)

From the NNO recursion principle one defines (uniquely) the standard maps

$$+ : N \times N \rightarrow N, \quad \times : N \times N \rightarrow N$$

by primitive recursion. For example, addition is characterized by:

$$a+0=a, \quad a+s(b)=s(a+b).$$

Then one proves (internally) the compatibility with iterators:

$$s^m(n)=n+m,$$

i.e. each iterator s^m is “add m ” once addition is defined.

Key takeaway:

- OFA(ii) gives you the **operator monoid** $\langle s \rangle$ immediately.
- Adding z and requiring the **recursion universal property** upgrades (N, z, s) to an **NNO**, from which full arithmetic is recoverable.

5. The conceptual package (one line)

OFA(ii) = the claim that the natural numbers are the **free monoid on one generator**, realized as the submonoid $\langle s \rangle \subseteq \text{End}(N)$, and that numerical “values” are obtained by evaluation at a basepoint z ; imposing the NNO recursion property makes (N, z, s) the canonical object supporting iteration/recursion in $\mathcal{C}$.

6. Fast falsifiability / validation checklist

1. **Monoid-theoretic check:** Verify that Φ_s is a monoid homomorphism with image $\langle s \rangle$.
2. **No-collapse check:** Determine whether $s^m = s^n \Rightarrow m = n$ holds in the intended model.
3. **NNO check:** With a chosen z , test the recursion property by constructing the unique u for arbitrary (A, q, f) and checking uniqueness.
4. **Cross-model invariance:** Compare a syntactic rewrite model and a semantic model (e.g. Sets/Types) and show they agree on closed terms via the initiality map.

References (external)

Core categorical foundations

- S. Mac Lane, *Categories for the Working Mathematician*, 2nd ed., Springer, 1998.
- S. Awodey, *Category Theory*, 2nd ed., Oxford University Press, 2010. (NNOs and recursion; also links to Lawvere theories.)
- T. Leinster, *Basic Category Theory*, Cambridge University Press, 2014. (Freely available as an arXiv preprint.)

Initiality, algebras, and Lambek's lemma

- J. Lambek, "A Fixpoint Theorem for Complete Categories," *Mathematische Zeitschrift* **103** (1968), 151–161. (Source of "Lambek's lemma" / initial algebra fixed-point property.)

Algebraic theories and universal constructions

- F. W. Lawvere, "Functorial Semantics of Algebraic Theories," *Proceedings of the National Academy of Sciences* **50** (1963), 869–872.

Handy online technical summaries

- nLab entries on natural numbers objects, free monoids, and initial algebras (good for quick cross-checks and references).
- Wikipedia pages on "natural numbers object" and "free monoid" (good for orientation and examples).

Convenience URLs (in a code block)

```
Lawvere (1963) PDF: https://www.sas.rochester.edu/mth/sites/doug-ravenel/otherpapers/lawvere.pdf
Lambek (1968) listing: https://eudml.org/doc/170906
nLab: natural numbers object: https://ncatlab.org/nlab/show/natural+numbers+object
nLab: free monoid: https://ncatlab.org/nlab/show/free+monoid
nLab: initial algebra of an endofunctor: https://ncatlab.org/nlab/show/
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initial+algebra+of+an+endofunctor

Wikipedia: NNO: https://en.wikipedia.org/wiki/Natural_numbers_object

Wikipedia: free monoid: https://en.wikipedia.org/wiki/Free_monoid

Leinster notes (arXiv): <https://arxiv.org/abs/1612.09375>

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